

FINA2204 Tutorial 8: Binomial Option Pricing Model

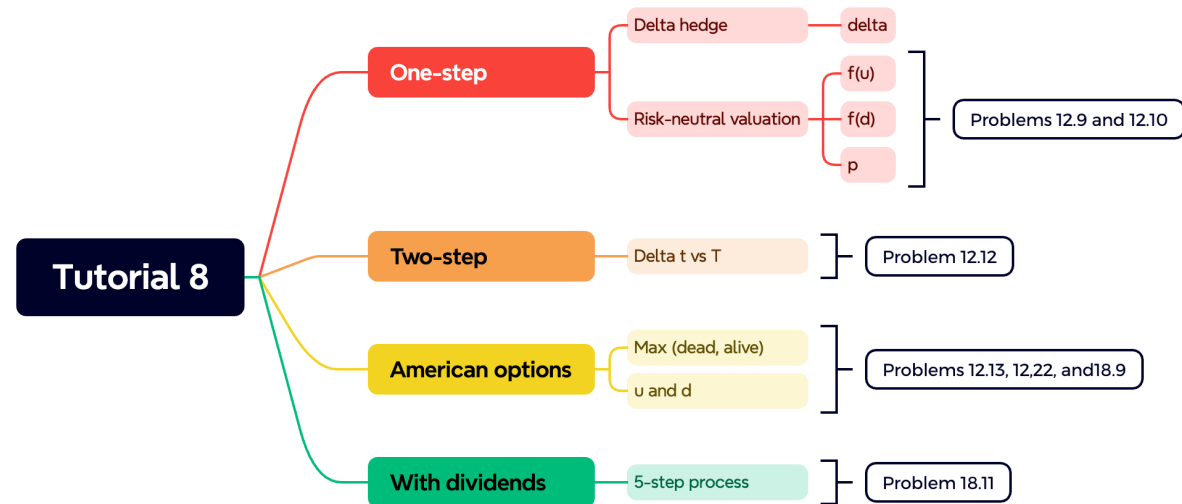
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Agenda

- Problem 12.9
- Problem 12.10
- Problem 12.12
- Problem 12.13
- Problem 12.22
- Problem 18.9
- Problem 18.11



Presented with xmind

Recap: one-step binomial model

- **Binomial tree**, created by Cox, Ross, and Rubinstein (CRR) in 1979, is a useful technique for **pricing an option (f_0)**.
- It is simply a way of representing possible future paths of the **stock price (S_T)**.

Known $S_T \rightarrow f_0$?

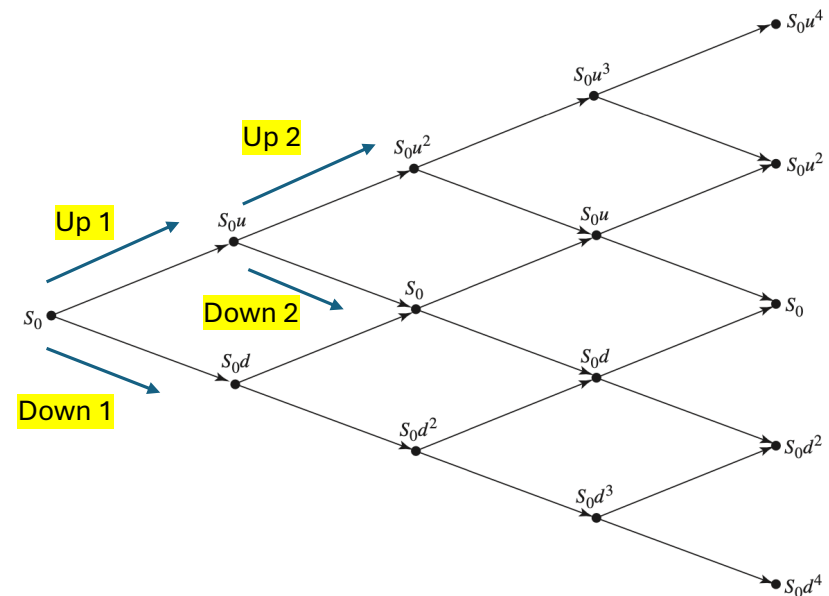


Figure 18.2 Tree used to value a stock option

Source: textbook, p.411

Recap: one-step binomial model

Setup: **Delta hedge approach**

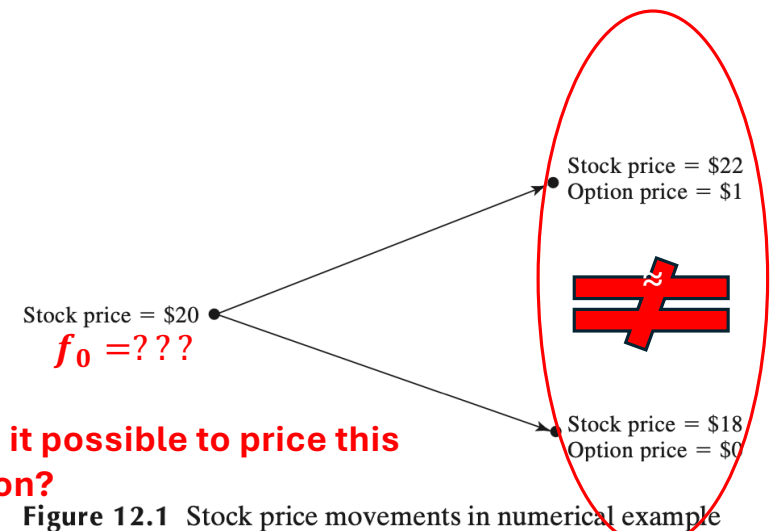
- Current stock price @ \$20
- $\pm 10\%$ change (@22 $\uparrow$, @18 $\downarrow$) in stock price over three months
- Strike price of a European call option @ \$21
- The risk-free rate is 12% per annum.

Two situations at maturity

1. if $S_T = 22$, then $c_T = (22 - 21) = 1$

2. if $S_T = 18$, then $c_T = 0$ (worthless)

- The option's payoff is **not risk-free**



Recap: one-step binomial model

Why not?

- In the real world, we would need two additional pieces of information.
- It depends on (i) whether the stock goes up or down (the probability of the stock going up, q), and (ii) because the option is risky, we would need an appropriate risk-adjusted discount rate.
- Neither of these is directly given to us. Both depend on beliefs about future returns and investors' risk appetites.

Recap: one-step binomial model

- Instead of relying on investors' subjective probabilities, we can leverage **no-arbitrage principles** in a risk-neutral world to construct a stock-and-option portfolio with a completely certain value at the end of three months.
- We then can argue that, since this future payoff is guaranteed, the risk-free portfolio must earn the risk-free interest rate.
- That gives us enough information to work backwards and determine the option price.

Analogy: Pricing the umbrella

- A Saturday picnic in Kings Park with an official ABC News forecast of 50% rain.
- You predict 70% sunshine; your friend expects a downpour.
- You tolerate a quick shower; your friend hates getting wet.
- **How do you agree on a fair price for the “umbrella option”?**



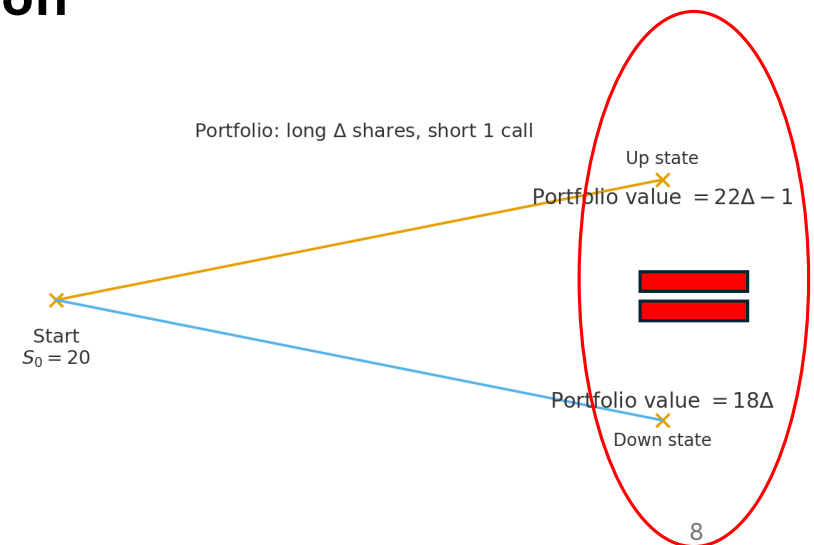
Analogy: Pricing the umbrella

- By acting like logical robots and trusting the objective ABC forecast, we find one mathematically fair price for the umbrella.



Recap: one-step binomial model

- Consider a portfolio consisting of a long position in Δ (“delta”) of the stock and a short position in one call option. We calculate the value of Δ that makes the portfolio riskless.
- So our portfolio is: **Δ stock - 1 call option**
- Our objective is to choose delta so that the final portfolio value is exactly the same in the up state and the down state.
- This means
$$22\Delta - 1 = 18\Delta \quad \text{or} \quad \Delta = 0.25$$



Recap: one-step binomial model

- A riskless portfolio is therefore
 1. Stock: +0.25 shares
 2. Call: -1 option
- If the stock price moves up to \$22, the value of the portfolio is
$$22 \times 0.25 - 1 = 4.5$$
- If the stock price moves down to \$18, the value of the portfolio is
$$18 \times 0.25 = 4.5$$

Regardless of whether the stock price moves up or down, the value of the portfolio is always 4.5 at the end of the life of the option.

Recap: one-step binomial model

- Because it is risk-free, its value today must be the present value of 4.50 discounted at the continuously compounded risk-free rate of 12 percent.

$$4.5e^{-0.12 \times 3/12} = 4.367$$

- The stock price today is known to be \$20. Suppose the option price is denoted by f . The value of the portfolio today is

$$20 \times 0.25 + f \times (-1) = 4.367$$

Or

$$f = \mathbf{0.6331}$$

Recap: one-step binomial model

Case 1: if $f > 0.6331$ (option is overpriced)

- Say $f = 0.80$.
- Then the portfolio value today is:
$$20 \times 0.25 + 0.80 \times (-1) = 4.20 < 4.367$$
- But it will be worth 4.5 later for sure.
- Strategy: borrow \$4.2 @ 12%; buy the portfolio.
- Results: the risk-free profit = $4.5 - 4.2 \times e^{0.12 \times \frac{3}{12}} = \0.18

Recap: one-step binomial model

Case 2: if $f < 0.6331$ (option is underpriced)

- Say $f = 0.50$.
- Then the portfolio value today is:
$$5 - 0.50 = 4.50 > 4.367$$
- Strategy: short sell the portfolio; Buy bond (\$4.5) @12%
- Results: the risk-free profit = $4.5 \times e^{0.12 \times \frac{3}{12}} - 4.5 = \0.14
- Therefore, both directions lead us back to the same no-arbitrage price.

Both cases confirm 0.6331 as the fair price of this option in the absence of arbitrage!

Recap: one-step binomial model

A generalization: **Risk-neutral valuation approach**

u : up factor

d : down factor

We can generalize the no-arbitrage argument just presented by considering a stock whose price is S_0 and an option on the stock (or any derivative dependent on the stock) whose current price is f . We suppose that the option lasts for time T and that during the life of the option the stock price can either move up from S_0 to a new level, S_0u , or down from S_0 to a new level, S_0d ($u > 1$; $d < 1$). The proportional increase in the stock price when there is an up movement is $u - 1$; the proportional decrease when there is a down movement is $1 - d$. If the stock price moves up to S_0u , we suppose that the payoff from the option is f_u ; if the stock price moves down to S_0d , we suppose the payoff from the option is f_d . The situation is illustrated in Figure 12.2.

Source: textbook, p.288

As before, we imagine a portfolio consisting of a long position in Δ shares and a short position in one option. We calculate the value of Δ that makes the portfolio riskless. If there is an up movement in the stock price, the value of the portfolio at the end of the life of the option is

$$S_0 u \Delta - f_u$$

If there is a down movement in the stock price, the value becomes

$$S_0 d \Delta - f_d$$

The two are equal when

$$S_0 u \Delta - f_u = S_0 d \Delta - f_d$$

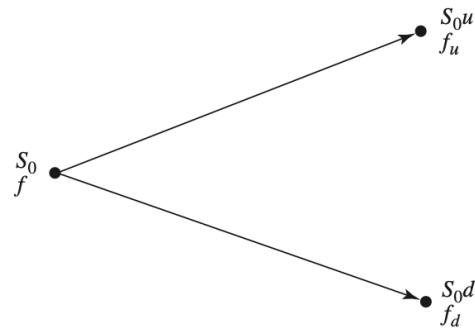


Figure 12.2 Stock and option prices in a general one-step tree

or

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d} \quad (12.1)$$

In this case, the portfolio is riskless and for there to be no arbitrage opportunities it must earn the risk-free interest rate. Equation (12.1) shows that Δ is the ratio of the change in the option price to the change in the stock price as we move between the nodes at time T .

If we denote the risk-free interest rate by r , the present value of the portfolio is

$$(S_0 u \Delta - f_u) e^{-rT}$$

The cost of setting up the portfolio is

$$S_0 \Delta - f$$

It follows that

$$S_0 \Delta - f = (S_0 u \Delta - f_u) e^{-rT}$$

or

$$f = S_0 \Delta (1 - u e^{-rT}) + f_u e^{-rT}$$

Substituting from equation (12.1) for Δ , we obtain

$$f = S_0 \left(\frac{f_u - f_d}{S_0 u - S_0 d} \right) (1 - u e^{-rT}) + f_u e^{-rT}$$

or

$$f = \frac{f_u (1 - d e^{-rT}) + f_d (u e^{-rT} - 1)}{u - d}$$

or

$$f = e^{-rT} [p f_u + (1 - p) f_d] \quad (12.2)$$

where

$$p = \frac{e^{rT} - d}{u - d} \quad (12.3)$$

A risk-neutral world has two features that simplify the pricing of derivatives:

1. The expected return on a stock (or any other investment) is the risk-free rate.
2. The discount rate used for the expected payoff on an option (or any other instrument) is the risk-free rate.

r

Returning to equation (12.2), the parameter p should be interpreted as the probability of an up movement in a risk-neutral world, so that $1 - p$ is the probability of a down movement in this world. (We assume that $u > e^{rT}$, so that $0 < p < 1$.) The expression

$$pf_u + (1 - p)f_d$$

is the expected future payoff from the option in a risk-neutral world and equation (12.2) states that the value of the option today is its expected future payoff in a risk-neutral world discounted at the risk-free rate. This is an application of risk-neutral valuation.

To prove the validity of our interpretation of p , we note that, when p is the probability of an up movement, the expected stock price $E(S_T)$ at time T is given by

$$E(S_T) = pS_0u + (1 - p)S_0d$$

or

$$E(S_T) = pS_0(u - d) + S_0d$$

Substituting from equation (12.3) for p gives

$$E(S_T) = S_0e^{rT} \quad (12.4)$$

This shows that the stock price grows, on average, at the risk-free rate when p is the probability of an up movement. In other words, the stock price behaves exactly as we would expect it to behave in a risk-neutral world when p is the probability of an up movement.

Source: textbook,
p.290-291

Recap: One-step binomial model

- Following equation (12.2) and (12.3),

$$f = e^{-rT} [pf_u + (1-p)f_d] \quad (12.2)$$

where

$$p = \frac{e^{rT} - d}{u - d} \quad (12.3)$$

$$u = \frac{22}{20} = 1.1, d = \frac{18}{20} = 0.9, p = \frac{e^{0.12 \times \frac{3}{12}} - 0.9}{1.1 - 0.9} = 0.6523$$
$$f = e^{-0.12 \times \frac{3}{12}} (0.6523 \times 1 + (1 - 0.6523) \times 0) = \mathbf{0.6331}$$

Real World vs. Risk-Neutral World

It should be emphasized that p is the probability of an up movement in a risk-neutral world. In general this is not the same as the probability of an up movement in the real world. In our example $p = 0.6523$. When the probability of an up movement is 0.6523, the expected return on both the stock and the option is the risk-free rate of 12%. Suppose that in the real world the expected return on the stock is 16%, and q is the probability of an up movement in the real world. It follows that

$$22q + 18(1 - q) = 20e^{0.16 \times 3/12}$$

so that $q = 0.7041$. **v.s. $p = 0.6523$**

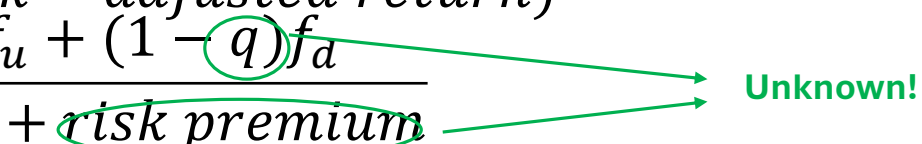
The expected payoff from the option in the real world is then

$$q \times 1 + (1 - q) \times 0$$

or 0.7041. Unfortunately, it is not easy to know the correct discount rate to apply to the expected payoff in the real world. The return the market requires on the stock is 16% and this is the discount rate that would be used for the expected cash flows from an investment in the stock. A position in a call option is riskier than a position in the stock. As a result, the discount rate to be applied to the payoff from a call option is greater than 16%, but we do not know how much greater than 16% it should be.¹ Using risk-neutral valuation is convenient because we know that in a risk-neutral world the expected return on all assets (and therefore the discount rate to use for all expected payoffs) is the risk-free rate.

Recap: One-step binomial model

In real-world,

$$\begin{aligned} f_0 &= \frac{E_{real}[f_T]}{(1 + \text{risk-adjusted return})} \\ &= \frac{qf_u + (1 - q)f_d}{1 + r + \text{risk premium}} \end{aligned}$$


Unknown!

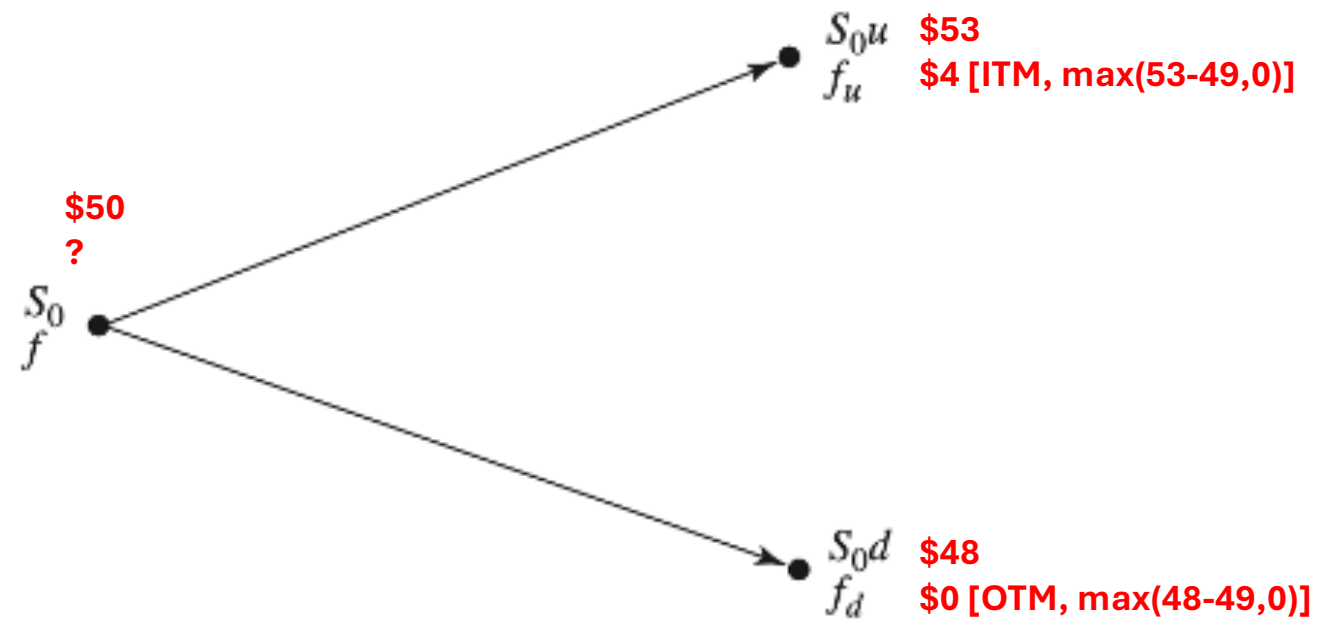
In risk-neutral world,

$$\begin{aligned} f_0 &= \frac{E_{risk-neutral}[f_T]}{(1 + \text{risk-free return})} \\ &= \frac{pf_u + (1 - p)f_d}{1 + r} \end{aligned}$$

Problem 12.9

- *A stock price is currently \$50. It is known that at the end of two months it will be either \$53 or \$48. The risk-free interest rate is 10% per annum with continuous compounding. What is the value of a two-month **European call option** with a strike price of \$49? Use no-arbitrage arguments.*

Problem 12.9



Problem 12.9

Method one: Delta hedge approach

- At the end of two months the value of the option will be either \$4 (if the stock price is \$53) or \$0 (if the stock price is \$48). Consider a portfolio consisting of:

+ Δ : *shares*

−1: *option*

The value of the portfolio is either 48Δ or $53\Delta - 4$ in two months. If

$$48\Delta = 53\Delta - 4$$

i.e., $\Delta = 0.8$

Problem 12.9

- The value of the portfolio is certain to be 38.4. For this value of riskless. The current value of the portfolio is:

$$0.8 \times 50 - f$$

where f is the value of the option. Since the portfolio must earn the risk-free rate of interest

$$(0.8 \times 50 - f)e^{0.10 \times \frac{2}{12}} = 38.4$$

i.e., $f = 2.23$

- The value of the option is therefore \$2.23.

Problem 12.9

Method two: Risk-neutral valuation approach

$$f = e^{-rT} [pf_u + (1-p)f_d] \quad (12.2)$$

where

$$p = \frac{e^{rT} - d}{u - d} \quad (12.3)$$

- $u = \frac{53}{50} = 1.06$ and $d = \frac{48}{50} = 0.96$. Thus
- $p = \frac{e^{0.10 \times 2/12} - 0.96}{1.06 - 0.96} = 0.5681$
- The value of call option is
- $f = e^{-0.10 \times \frac{2}{12}} (0.5681 \times 4 + 0.4319 \times 0) = \2.23

Recap: One-step binomial model

Pricing put options

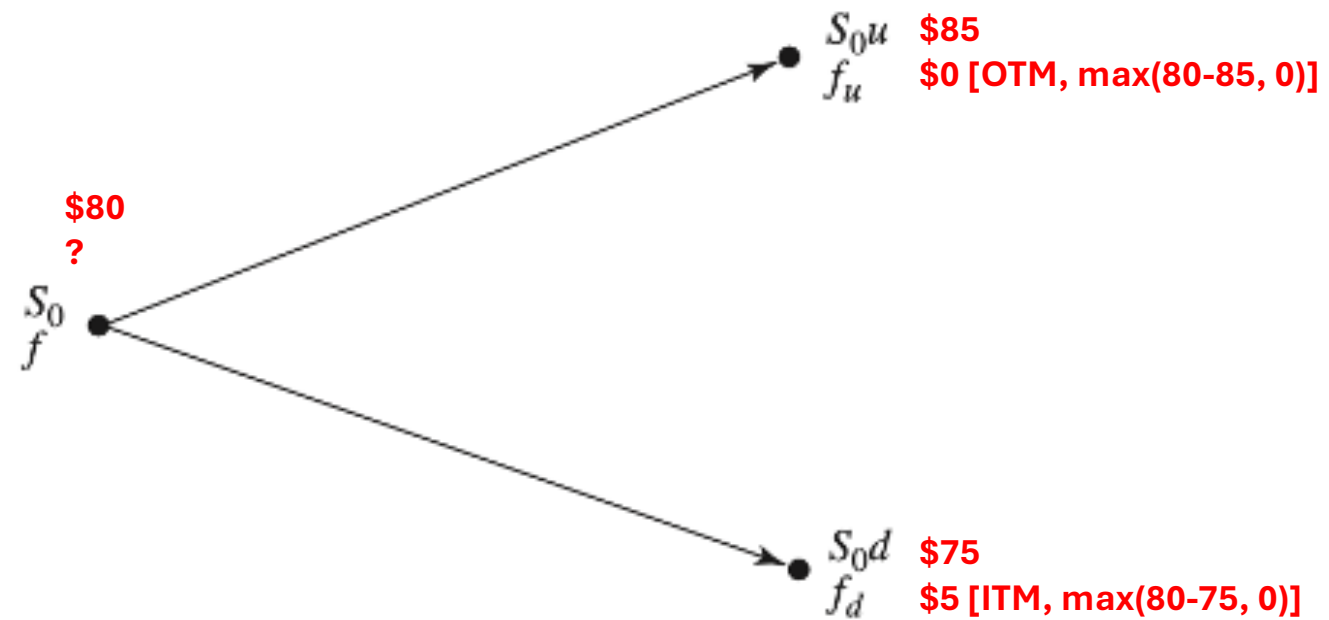
- The binomial framework itself **does not change**: (i) the underlying stock price tree (S_0, u, d) is identical; (2) the risk-neutral probability (p) is calculated using the exact same formula.
- The key difference: Because a put becomes more valuable when the stock price falls, the put's delta (Δ) is **negative**.
- Thus, we set up the portfolio as:
 - $-\Delta$ shares
 - $+1$ put option @ f

In this case, the portfolio is riskless and for there to be no arbitrage opportunities it must earn the risk-free interest rate. Equation (12.1) shows that Δ is the ratio of the change in the option price to the change in the stock price as we move between the nodes at time T . If we denote the risk-free interest rate by r , the present value of the portfolio is

Problem 12.10

- *A stock price is currently \$80. It is known that at the end of four months it will be either \$75 or \$85. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a four-month **European put option** with a strike price of \$80? Use no-arbitrage arguments.*

Answer 12.10



Answer 12.10

Method one: Delta hedge approach

- At the end of four months the value of the option will be either \$5 (if the stock price is \$75) or \$0 (if the stock price is \$85). Consider a portfolio consisting of:
- $-\Delta$ shares
- +1 put option @ f

Note: the delta of a put option is negative. We have constructed the portfolio so that it is +1 option and $-\Delta$ shares rather than -1 option and $+\Delta$ shares so that the initial investment is positive.

Answer 12.10

- The value of the portfolio is either -85Δ or $-75\Delta + (80 - 75)$ in four months. If

$$-85\Delta = -75\Delta + 5$$

(i.e. $\Delta = -0.5$). The value of the portfolio is certain to be 42.5. For this value of Δ , the portfolio is therefore riskless. The current value of the portfolio is

$$0.5 \times 80 + f$$

where f is the value of the option. Since the portfolio is riskless

$$(0.5 \times 80 + f)e^{0.05 \times \frac{4}{12}} = 42.5$$

i.e. $f = \$1.80$. The value of the option is therefore **\$1.80**.

Answer 12.10

Method two: Risk-neutral valuation approach

$$f = e^{-rT} [pf_u + (1-p)f_d] \quad (12.2)$$

where

$$p = \frac{e^{rT} - d}{u - d} \quad (12.3)$$

This can also be calculated directly from equation (12.2) and (12.3).

Since $u = \frac{85}{80} = 1.0625$, and $d = \frac{75}{80} = 0.9375$

$$p = \frac{e^{0.05 \times 4/12} - 0.9375}{1.0625 - 0.9375} = 0.6345$$

And $1 - p = 0.3655$.

- The value of the put option is

$$f = e^{-0.05 \times \frac{4}{12}} (0.6345 \times 0 + 0.3655 \times 5) = \$1.80$$

Recap: Two-step binomial model

- **Setup (extended)**: Here the stock price starts at \$20 and in each of **two time steps** may go up by 10% or down by 10%. Each time step is three months long and the risk-free interest rate is 12% per annum. We consider a six-month call option with a strike price of \$21.
- A useful feature of this tree is that it **recombines**: an up move followed by a down move reaches the same node as a down move followed by an up move ($S_0 * u * d = S_0 * d * u$).
- And importantly, the risk-neutral probability (p) is **constant** at every step.

← The two-step binomial tree is evaluated backward, from right to left.

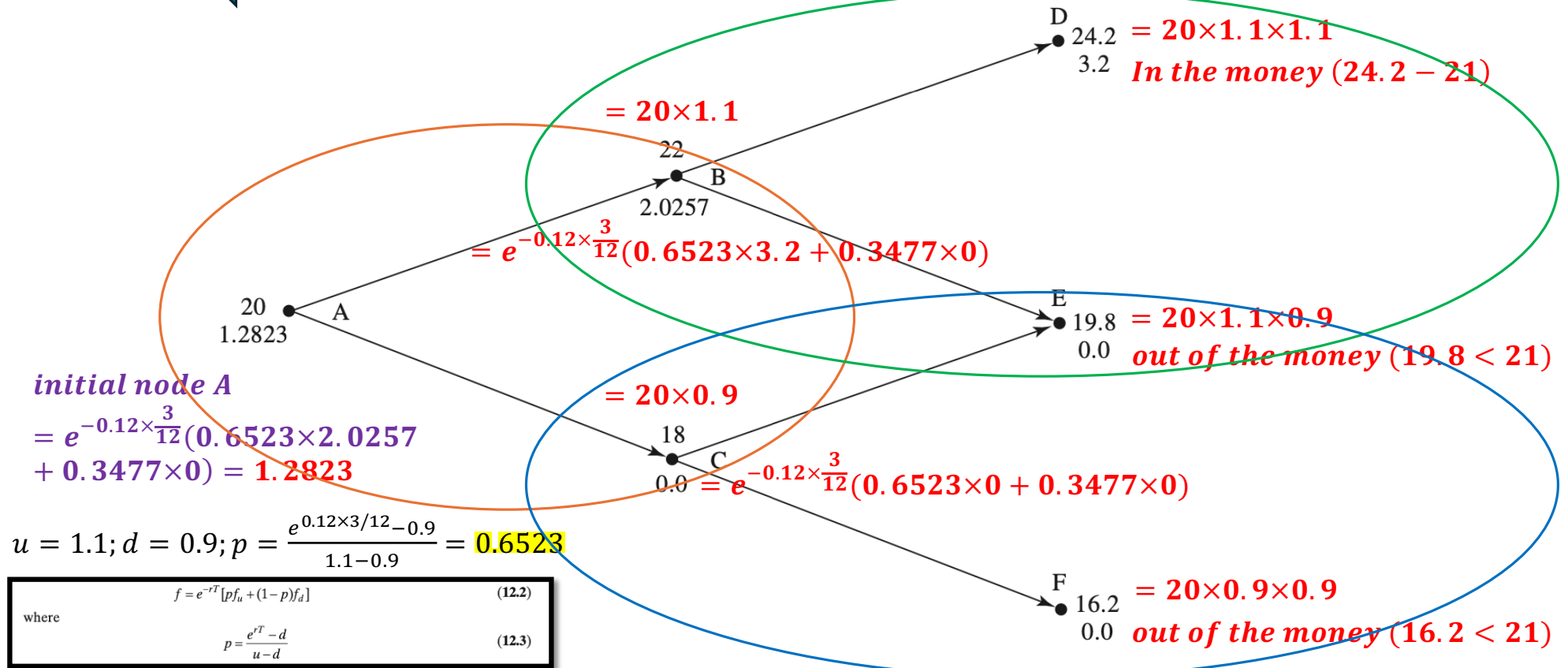


Figure 12.4 Stock and option prices in a two-step tree. The upper number at each node is the stock price; the lower number is the option price

Recap: Two-step binomial model

We can generalize the case of two time steps by considering the situation in Figure 12.6. The stock price is initially S_0 . During each time step, it either moves up to u times its

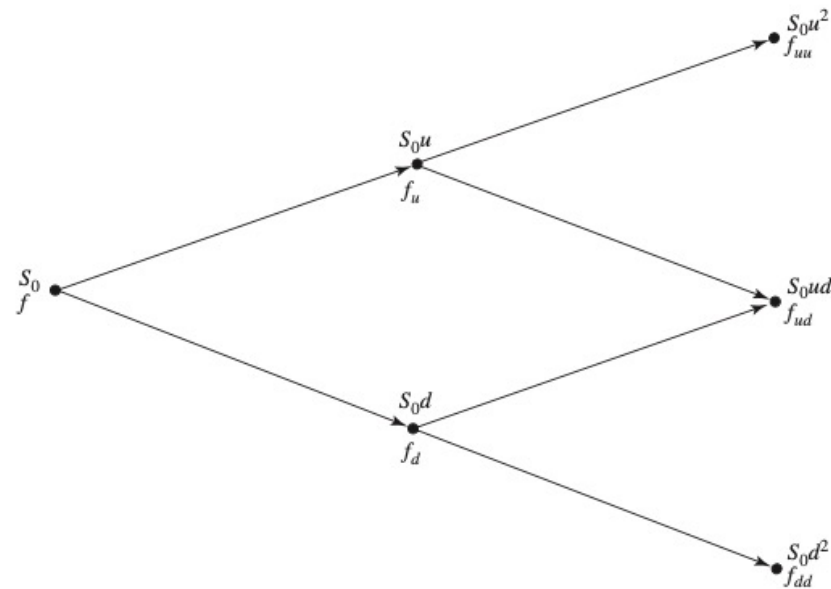


Figure 12.6 Stock and option prices in general two-step tree

initial value or moves down to d times its initial value. The notation for the value of the option is shown on the tree. (For example, after two up movements the value of the option is f_{uu} .) We suppose that the risk-free interest rate is r and the length of the time step is Δt years.

Because the length of a time step is now Δt rather than T , equations (12.2) and (12.3) become

$$f = e^{-r\Delta t} [pf_u + (1-p)f_d] \quad (12.5)$$

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (12.6)$$

Δt vs. T

Δt is the length of **one time** step on the tree.

T is the **total time** to maturity (sum of all steps).

Repeated application of equation (12.5) gives

$$f = e^{-rT} [pf_u + (1-p)f_d] \quad (12.2)$$

$$p = \frac{e^{rT} - d}{u - d} \quad (12.3)$$

$$f_u = e^{-r\Delta t} [pf_{uu} + (1-p)f_{ud}] \quad (12.7)$$

$$f_d = e^{-r\Delta t} [pf_{ud} + (1-p)f_{dd}] \quad (12.8)$$

$$f = e^{-r\Delta t} [pf_u + (1-p)f_d] \quad (12.9)$$

Substituting from equations (12.7) and (12.8) into (12.9), we get

$$f = e^{-2r\Delta t} [p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}] \quad (12.10)$$

This is consistent with the principle of risk-neutral valuation mentioned earlier. The variables p^2 , $2p(1-p)$, and $(1-p)^2$ are the probabilities that the upper, middle, and lower final nodes will be reached. The option price is equal to its expected payoff in a risk-neutral world discounted at the risk-free interest rate.

As we add more steps to the binomial tree, the risk-neutral valuation principle continues to hold. The option price is always equal to its expected payoff in a risk-neutral world, discounted at the risk-free interest rate.

Only for the
European
options!

Source: textbook,
p.295

Recap: Two-step binomial model

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (12.6)$$

- Following equation (12.6), $u = 1.1$; $d = 0.9$; $p = \frac{e^{0.12 \times 3/12} - 0.9}{1.1 - 0.9} = 0.6523$

- And $f_{uu} = 3.2$, $f_{ud} = 0$, $f_{dd} = 0$

- Following equation $f = e^{-2r\Delta t} [p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}] \quad (12.10)$

$$f = e^{-2 \times 0.12 \times \frac{3}{12}} (0.6523^2 \times 3.2 + 2 \times 0.6523 \times (1 - 0.6523) \times 0 + (1 - 0.6523)^2 \times 0) = \mathbf{1.2823}$$

Recap: Two-step binomial model

A put example

- The procedures described before can be used to price puts as well as calls.

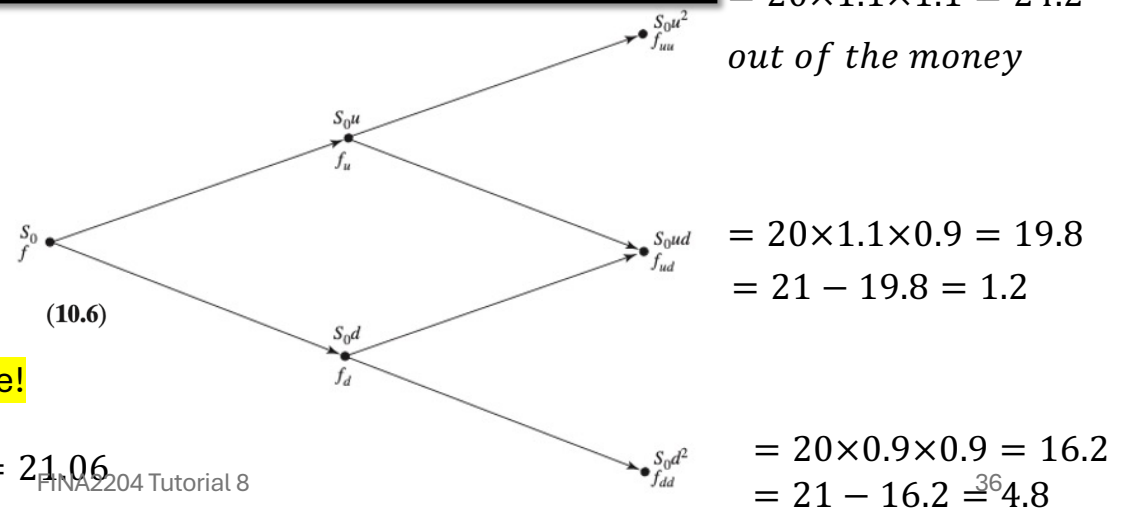
$$f = e^{-2r\Delta t} [p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}] \quad (12.10)$$

$$f = e^{-2 \times 0.12 \times \frac{3}{12}} (0.6523^2 \times 0 + 2 \times 0.6523 \times (1 - 0.6523) \times 1.2 + (1 - 0.6523)^2 \times 4.8) = 1.06$$

$$c + Ke^{-rT} = p + S_0$$

Put-call parity is satisfied in this example!

$$1.2823 + 21 \times e^{-0.12 \times 6/12} = 21.06 \quad \blacksquare \quad 1.06 + 20 = 21.06$$



Problem 12.12

- A stock price is currently \$50. Over **each of the next two three-month** periods it is expected to go up by 6% or down by 5%. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a six-month **European call option** with a strike price of \$51?

Answer 12.12

- Following equation (12.6)
$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (12.6)$$
- The risk-neutral probability of an up move, p , is given by
$$p = \frac{e^{0.05 \times 3/12} - 0.95}{1.06 - 0.95} = 0.5689$$
- There is a payoff from the option of $50 \times 1.06 \times 1.06 - 51 = 56.18 - 51 = 5.18$ for the highest final node (f_{uu}). And Zero in other two cases (out of the money, $50 \times 0.95 \times 0.95 < 50 \times 1.06 \times 0.95 < 51$).

Approach one

- It can be calculated by working back through tree as indicated in Figure S12.1.

Answer 12.12

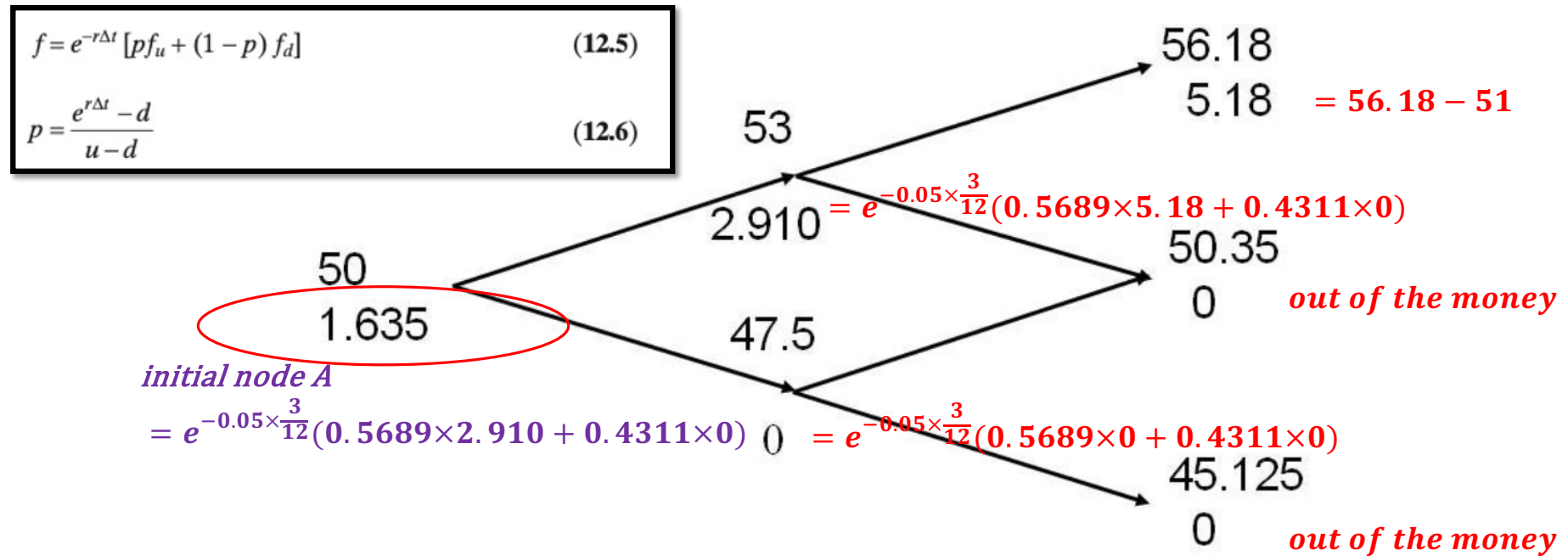


Figure S12.1: Tree for Problem 12.12

Answer 12.12

Approach two

- Using equation $f = e^{-2r\Delta t} [p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}]$ (12.10)

- the value of the option is therefore

$$f = e^{-2 \times 0.05 \times \frac{3}{12}} (0.5689^2 \times 5.18 + 2 \times 0.5689 \times 0 + (1 - 0.5689)^2 \times 0) \\ = 1.635$$

Recap: American options

Up to now all the options we have considered have been European. We now move on to consider how American options can be valued using binomial trees such as those in Figures 12.4 and 12.7. The procedure is to work back through the tree from the end to the beginning, testing at each node to see whether early exercise is optimal. The value of the option at the final nodes is the same as for the European option. At earlier nodes the value of the option is the greater of:

1. The value given by equation (12.5)
2. The payoff from early exercise.

$$f = e^{-r\Delta t} [pf_u + (1 - p) f_d] \quad (12.5)$$

Call: $\max(S - K, 0)$; *Put:* $\max(K - S, 0)$

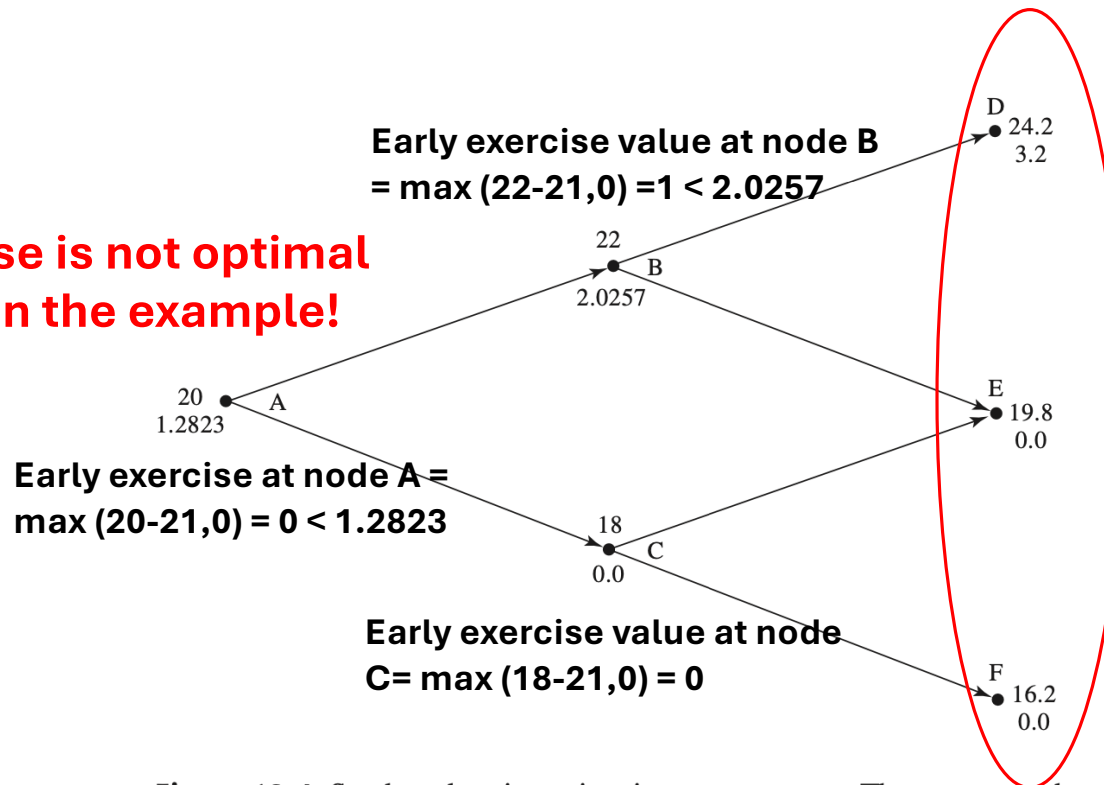
Source: textbook, p.296

Rule: Value of the American option = **Max (alive, dead)**; If dead is greater than alive, exercise early.

Recap: American options

Setup (extended): Here the stock price starts at \$20 and in each of two time steps may go up by 10% or down by 10%. Each time step is three months long and the risk-free interest rate is 12% per annum. We consider a six-month call option with a **strike price of \$21**.

Early exercise is not optimal anywhere in the example!



At the final nodes, there is no more time remaining, so American and European values are identical.

The early-exercise comparison begins one step before maturity.

Figure 12.4 Stock and option prices in a two-step tree. The upper number at each node is the stock price; the lower number is the option price

Problem 12.13

- *For the situation considered in Problem 12.12, what is the value of a six-month **European put option** with a strike price of \$51? Verify that the European call and European put prices satisfy put–call parity. If the put option were **American**, would it ever be optimal to exercise it early at any of the nodes on the tree?*
- *(Problem 12.12: A stock price is currently \$50. Over **each of the next two three-month** periods it is expected to go up by 6% or down by 5%. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a six-month **European call option** with a strike price of \$51?)*

Value of the call → 1.635

Answer 12.13

$$f = e^{-r\Delta t} [pf_u + (1 - p) f_d] \quad (12.5)$$

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (12.6)$$

Inputs

Six-month **European put option**

- $u = 1.06, d = 0.95$
- $S_0 = 50, K = 51$
- $r = 5\%$
- $\Delta t = 3/12$
- $p = \frac{e^{0.05 \times 3/12} - 0.95}{1.06 - 0.95} = 0.5689$

Answer 12.13

$$f = e^{-r\Delta t} [pf_u + (1-p)f_d] \quad (12.5)$$

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (12.6)$$

- $f_{uu} = \max(51 - 56.18, 0) = 0$
- $f_{ud} = \max(51 - 50.35, 0) = 0.65$
- $f_{ud} = \max(51 - 45.125, 0) = 5.875$
- $f_u = e^{-0.05 \times 3/12} [0.5689 \times 0 + 0.4311 \times 0.65] = 0.277$
- $f_d = e^{-0.05 \times 3/12} [0.5689 \times 0.65 + 0.4311 \times 5.875] = 2.866$
- $f_0 = e^{-0.05 \times 3/12} [0.5689 \times 0.277 + 0.4311 \times 2.866] = 1.376$

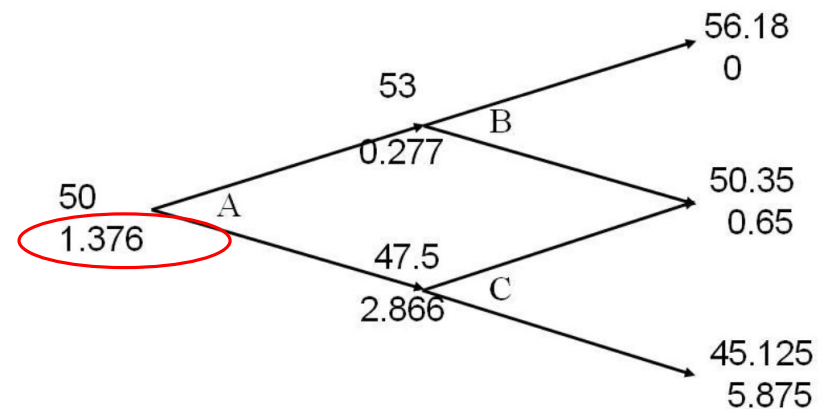


Figure S12.2 Tree for Problem 12.13

Answer 12.13

- Following equation (12.10), the value of the put option is therefore

$$f = e^{-2r\Delta t} [p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}] \quad (12.10)$$

- $f = e^{-2 \times 0.05 \times \frac{3}{12}} (2 \times 0.5689 \times 0.4311 \times 0.65 + 0.4311^2 \times 5.875) =$
1.376

$$c + Ke^{-rT} = p + S_0 \quad (10.6)$$

$$1.635 + 51e^{-0.05 \times 6/12} = 51.376$$

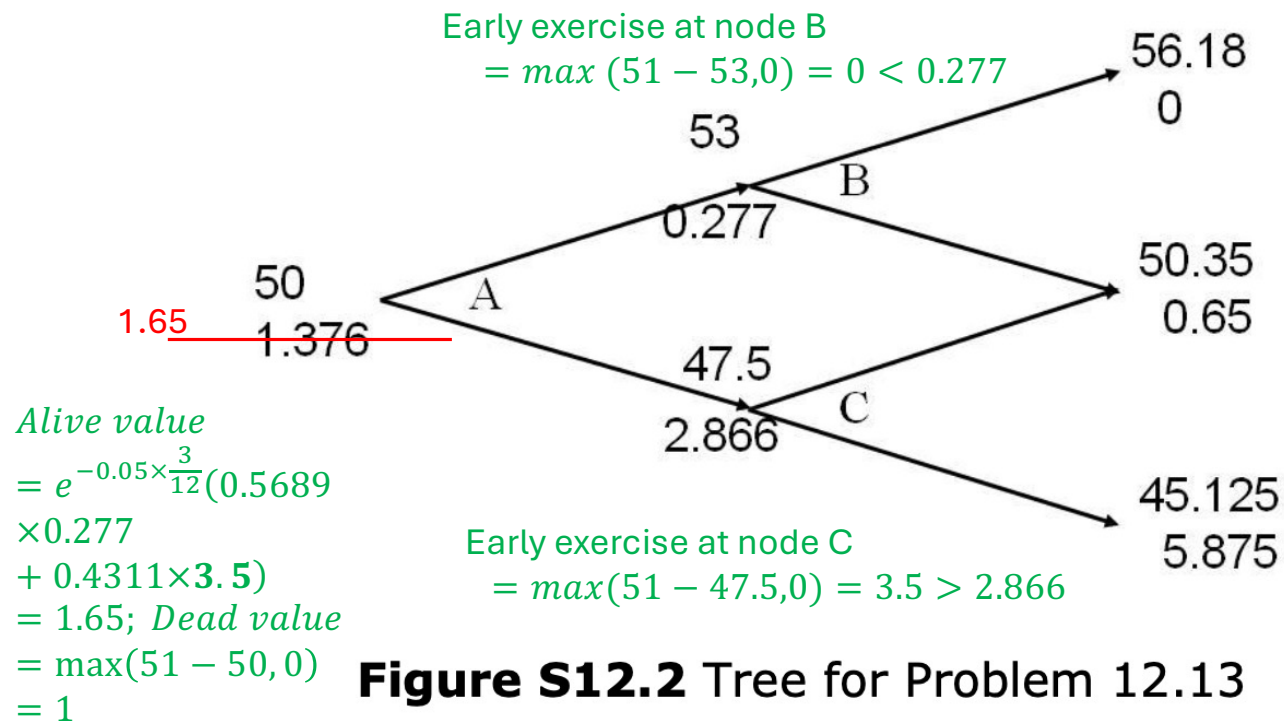
$$1.376 + 50 = 51.376$$

- This verifies that put-call parity holds.

Answer 12.13

- To test whether it is worth exercising the option early, we compare the value calculated for the option at each node with the payoff from immediate exercise.
- At **node C** the payoff from immediate exercise is $51 - 47.5 = 3.5$. Because this is greater than 2.8664, the option **should be exercised at this node**.
- The option should not be exercised at either node A or node B.

Answer 12.13



At the final nodes, there is no more time remaining, so American and European values are identical.

The early-exercise comparison begins one step before maturity.

Problem 12.22

- *A stock price is currently \$40. Over each of the next two three-month periods it is expected to go up by 10% or down by 10%. The risk-free interest rate is 12% per annum with continuous compounding.*
- *a. What is the value of a six-month European put option with a strike price of \$42?*
- *b. What is the value of a six-month American put option with a strike price of \$42?*

Answer 12.22

$$f = e^{-r\Delta t} [pf_u + (1 - p) f_d] \quad (12.5)$$

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (12.6)$$

Inputs

Six-month **European/American put option**

- $u = 1.1, d = 0.9$
- $S_0 = \$40, K = \42
- $\Delta t = 3/12$
- $r = 12\%$
- $p = \frac{e^{0.12 \times 3/12} - 0.9}{1.1 - 0.9} = 0.6523$

Answer 12.22

- A tree describing the behavior of the stock price is shown in Figure S12.7.
- Following equation (12.10),

$$f = e^{-2r\Delta t} [p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}] \quad (12.10)$$

- We obtain the value of the **European** put option as
$$f = e^{-2 \times 0.12 \times 3/12} [0.6523^2 \times 0 + 2 \times 0.6523 \times 0.3477 \times 2.4 + 0.3477^2 \times 9.6] = 2.118$$

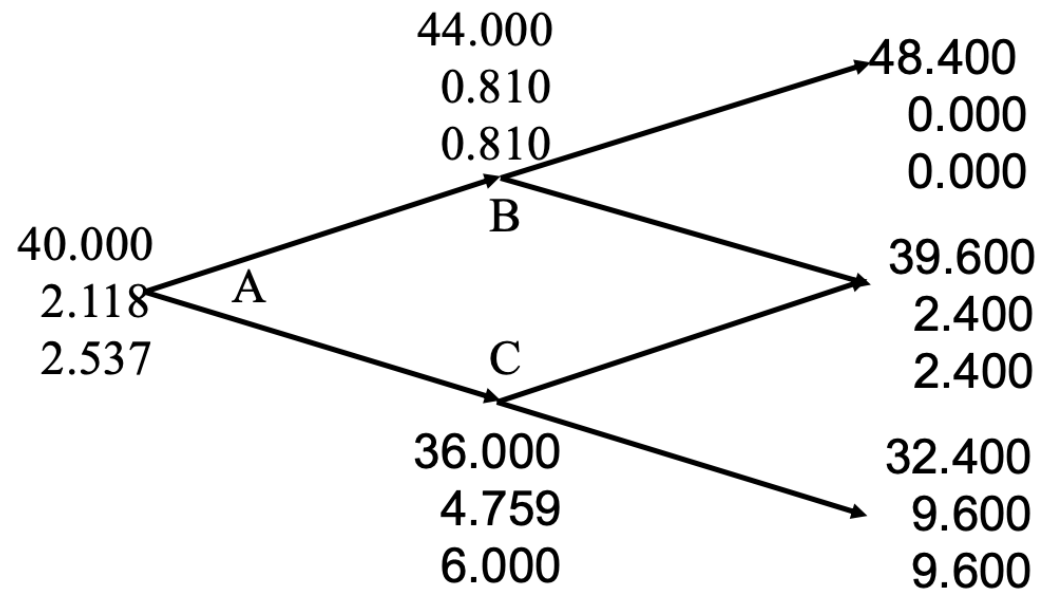
Answer 12.22

- The value of the European option is 2.118. This can also be calculated by working back through the tree as shown in Figure S12.7. The second number at each node is the value of the European option.
- The value of the American option is shown as the third number at each node on the tree. It is 2.537. This is greater than the value of the European option **because it is optimal to exercise early at node C.**

Answer 12.22

$$f = e^{-r\Delta t} [pf_u + (1 - p) f_d] \quad (12.5)$$

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (12.6)$$



The value given by equation (12.5) = $e^{-0.12 \times 3/12} [0.6523 \times 2.4 + 0.3477 \times 9.6] = 4.759$

2/10/2026 *Payoff from the early exercise at node C: $\max(K - S, 0) = \max(42 - 36, 0) = 6$*

Recap: Determining u and d

Up to now we have assumed values for u and d . For example, in Figure 12.8, we assumed that $u = 1.2$ and $d = 0.8$. In practice, u and d are determined from the stock price volatility, σ . The formulas are

$$u = e^{\sigma\sqrt{\Delta t}} \quad (12.11)$$

u and d are reciprocals of each other.

$$d = \frac{1}{u} \quad (12.12)$$

where Δt is the length of one time step on the tree. These formulas will be explained further in Chapter 18. The complete set of equations defining the tree are equation (12.11) and equation (12.12), together with equation (12.6), which can be written

$$p = \frac{a - d}{u - d} \quad (12.13)$$

where

$$a = e^{r\Delta t} \quad (12.14)$$

Problem 18.9

- *A nine-month American put option on a non-dividend-paying stock has a strike price of \$49. The stock price is \$50, the risk-free rate is 5% per annum, and the volatility is 30% per annum. Use a **three-step binomial tree** to calculate the option price.*

Answer 18.9

$$f = e^{-r\Delta t} [pf_u + (1 - p) f_d] \quad (12.5)$$

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (12.6)$$

Inputs

*Nine-month **American put** option*

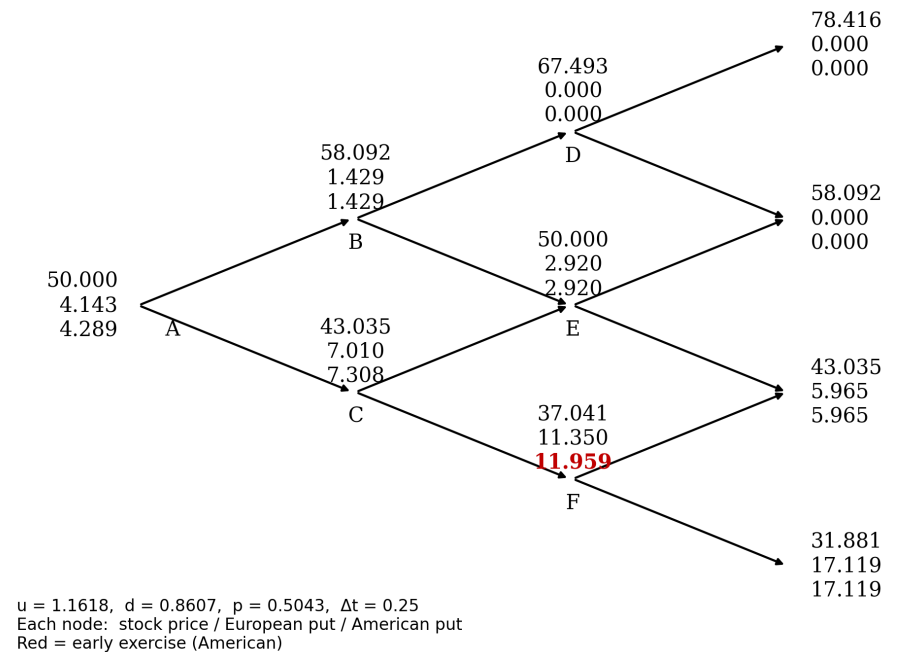
- $T = \frac{9}{12}, \Delta t = \frac{9}{12} \times \frac{1}{3} = 0.25$
- $\sigma = 0.30$
- $u = e^{0.30 \times \sqrt{0.25}} = 1.1618, d = \frac{1}{u} = 0.8607$
- $S_0 = 50, K = 49$
- $r = 0.05$
- $p = \frac{e^{-0.05 \times 0.25} - 0.8607}{1.1618 - 0.8607} = 0.5043$

$$u = e^{\sigma\sqrt{\Delta t}} \quad (12.11)$$

$$d = \frac{1}{u} \quad (12.12)$$

Answer 18.9

- The output from *DerivaGem* for this example is shown in the Figure S18.1. The calculated price of the option is \$4.29.
- Early exercise is optimal only **at node F** (dead 11.959 > alive 11.350), which lifts the American put to 4.289 vs European 4.143.



Recap: American options (with dividends)

- When a known dollar dividend is subtracted from the stock price at the ex-dividend date, the binomial tree **no longer recombines**.
- $(s_0u - D) \times d \neq (s_0d - D) \times u$
- A 2-step recombining tree has 3 terminal nodes; without recombination, it wrongly produces **4!**

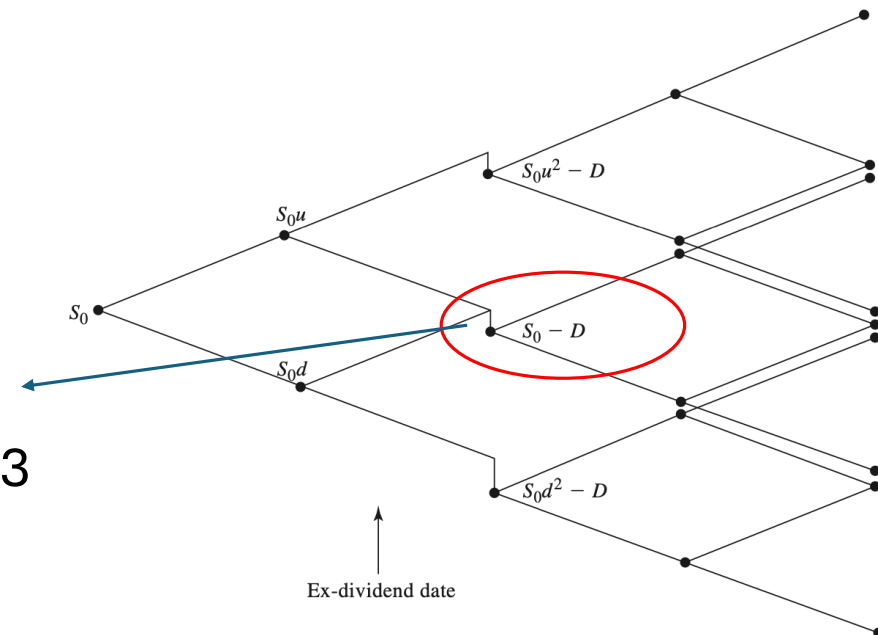


Figure 18.6 Tree when dollar amount of dividend is assumed known and volatility is assumed constant

Recap: American options (with dividends)

To reflect the stock's true value (dividends collected) if exercised early.

For European options

1. compute the PV of all dividend due to be paid prior to expiration
2. subtract PV of dividends from S to obtain S^*

$$S^* = S - \sum_{j=1}^m D_j e^{-rt_j}$$

3. construct stock price tree starting from S^*
4. apply binomial model to terminal payoffs as for non-dividend stocks

For American options

1. follow steps 1-3 as for European option
4. add back to stock price at each node the PV of all future dividends to be paid during the remaining life of option
5. compute option price by working backwards through tree. At each node:
 - calculate the immediate exercise value of the option (value "dead")
 - calculate the value of the option "alive" using the 1-period formula
 - choose the maximum of the "dead" and "alive" values as the value of the option at that node.

comparison

of the option at the final nodes is the same as for the European option. At earlier nodes the value of the option is the greater of:

1. The value given by equation (12.5)

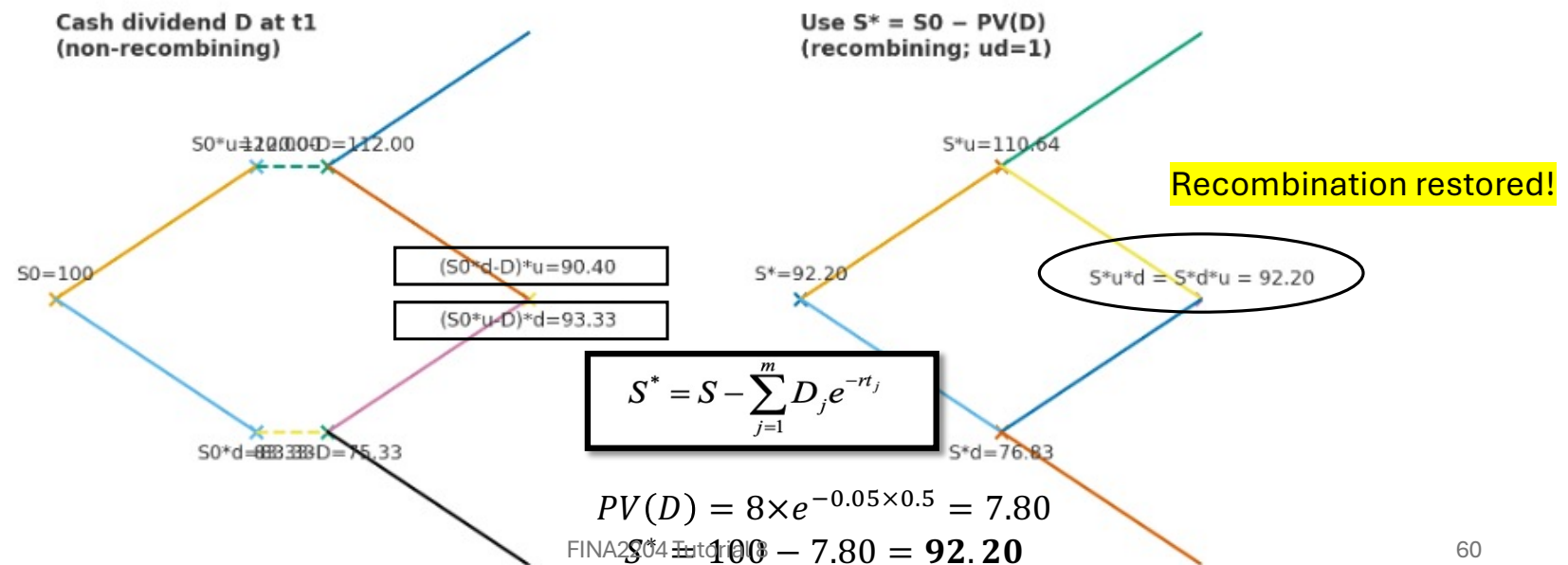
2/10/2026 2. The payoff from early exercise.

$$f = e^{-r\Delta t} [pf_u + (1 - p) f_d] \quad (12.5)$$

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Recap: American options (with dividends)

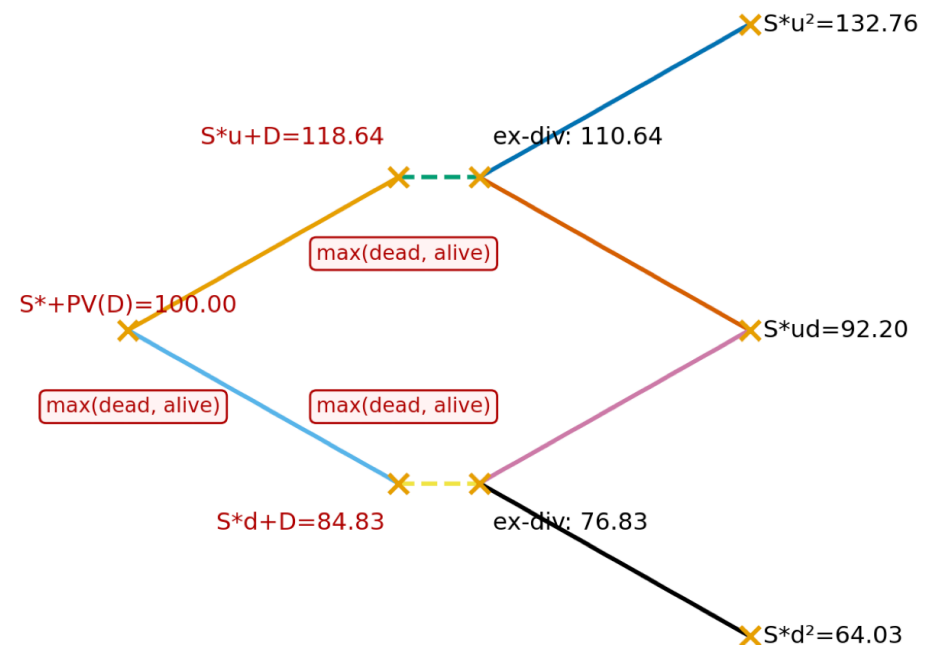
- For European options: Started at $S_0 = K = 100$, with $u = 1.2$, $d = \frac{1}{u}$, $D = 8$, $r = 5\%$, six-month period.



Recap: American options (with dividends)

- For American options:
Same setting.
- Add back PV(D) at each node before t, then take **max(dead, alive)** at each node.
- Conclusion: American 9.52 > European 8.54

Steps 4-5 (American): add back PV of future dividends, then roll back with **max(dead, alive)** at each node



Problem 18.11

- A three-month **American call option** on a stock has a strike price of \$20. The stock price is \$20, the risk-free rate is 3% per annum, and the volatility is 25% per annum. **A dividend** of \$2 is expected in 1.5 months. Use a **three-step binomial tree** to calculate the option price.

Answer 18.11

$$u = e^{\sigma\sqrt{\Delta t}} \quad (12.11)$$

$$d = \frac{1}{u} \quad (12.12)$$

- In this case the present value of the dividend is $2e^{-0.03 \times \frac{1.5}{12}} = 2e^{-0.03 \times 0.125} = 1.9925$.
- We first build a tree for $S_0 = 20 - 1.9925 = 18.0075$, $K = 20$, $r = 0.03$, $\sigma = 0.25$, and $T = \frac{3}{12}$ with $\Delta t = \frac{3}{12} \times \frac{1}{3} = 0.0833$.
- $u = e^{0.25\sqrt{0.08333}} = 1.0748$; $d = \frac{1}{u} = 0.9304$
- $p = \frac{e^{0.03 \times 0.0833} - 0.9304}{1.0748 - 0.9304} = 0.4993$
- This gives Figure S18.3.

Answer 18.11

- For nodes between times 0 and 1.5 months we then add the **present value** of the dividend to the stock price.
- The result is the tree in Figure S18.4.
- The price of the option calculated from the tree is **0.674**
[$= e^{-0.03 \times 0.0833} (1.352609 \times 0.4993 + 0 \times (1 - 0.4993))$].
- When **100 steps** are used the price obtained is 0.690 (*Note: the more steps you use, the shorter each time interval Δt becomes, and the more accurate the approximation of continuous price movements*).

